Planned Pooling Project Pack

Worksheets, three worked projects, and a dye-lot buying checklist for planned color pooling

A printable companion to the free Planned Color Pooling Calculator at FiberTools — the calculators stay free; this pack is the paper bridge from preview to finished piece.

How pooling math works here

A color repeat is the ordered whole-stitch counts of one full color sequence.

Row width = repeat + adjustment

Shift per row = row width mod repeat

Shift 0 stacks colors into columns in the sequence. Shift 1 steps the pattern one stitch per row (argyle drift); larger shifts step faster. These describe sequence offsets; the grid shows the turned fabric.

Turned rows are worked back across, so the fabric mirrors while the sequence keeps consuming in order.

Measure, preview, then swatch

Record one complete sequence, try nearby widths, and work a swatch at the planned width. Keep the trial that suits your real fabric and use its measured yarn consumption for your project plan.

Limits

Whole stitches only; color boundaries can land inside a stitch. No foundation-chain or turning-chain consumption is modeled. Tension, joins, edges, and repeat variation change real fabric. The grid is an idealized planning aid, not a guaranteed pooling outcome.

W1 — Yarn and Repeat Record

Yarn brand / line

Weight (CYC)

Dye lot

Skein yardage

Hook / needle

Color	Hex	Whole stitches

Total repeat (whole stitches)

W2 — Trial Width Log

Use a fresh swatch at each trial width and record what the fabric actually did.

Adjustment	Row width	Shift per row	What the swatch did	Keep?

W3 — Blank Pooling Grid

30 rows × 40 columns. Mark your colors and direction of work.

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Color key

Code / mark	Color	Hex

W4 — Project Plan and Yardage

Measured swatch

Rows worked

Yarn used (yards)

Yards per row = yarn used ÷ rows worked

Project

Planned rows (include all panels)

Total yards = yards per row × planned rows

Overage %

Yards to buy = total yards × (1 + overage % / 100)

Skein size (yards)

Skeins = round up (yards to buy ÷ skein size)

Dye lot

W5 — Row Tracker

Check each row as you finish it.

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Argyle Drift Scarf

Argyle Drift Scarf · Fingering (CYC 1) · crochet moss stitch

> Worked planning example: every stitch number below was computed with the FiberTools pooling model; the swatch yardage is an example measurement — replace it with your own swatch numbers.

Color	Name	Hex	Whole stitches
TE	Deep Teal	#0F5E63	6
CO	Coral	#E2725B	4
CR	Cream	#F2E8C9	6
PL	Plum	#6C3483	4

Total repeat: 20 stitches

Adjustment	Row width	Shift per row
-2	18	18
-1	19	19
0	20	0
1	21	1
2	22	2

Chosen adjustment: **1**; row width: 21; shift per row: 1. A one-stitch step makes the argyle drift.

Expected pattern grid

1	TE	TE	TE	TE	TE	TE	CO	CO	CO	CO	CR	CR	CR	CR	CR	CR	PL	PL	PL	PL	TE
2	TE	TE	PL	PL	PL	PL	CR	CR	CR	CR	CR	CR	CO	CO	CO	CO	TE	TE	TE	TE	TE
3	TE	TE	TE	TE	CO	CO	CO	CO	CR	CR	CR	CR	CR	CR	PL	PL	PL	PL	TE	TE	TE
4	TE	TE	TE	TE	PL	PL	PL	PL	CR	CR	CR	CR	CR	CR	CO	CO	CO	CO	TE	TE	TE
5	TE	TE	CO	CO	CO	CO	CR	CR	CR	CR	CR	CR	PL	PL	PL	PL	TE	TE	TE	TE	TE
6	TE	TE	TE	TE	TE	TE	PL	PL	PL	PL	CR	CR	CR	CR	CR	CR	CO	CO	CO	CO	TE
7	CO	CO	CO	CO	CR	CR	CR	CR	CR	CR	PL	PL	PL	PL	TE	TE	TE	TE	TE	TE	CO
8	CO	CO	TE	TE	TE	TE	TE	TE	PL	PL	PL	PL	CR	CR	CR	CR	CR	CR	CO	CO	CO
9	CO	CO	CR	CR	CR	CR	CR	CR	PL	PL	PL	PL	TE	TE	TE	TE	TE	TE	CO	CO	CO

10	CO	CO	CO	CO	TE	TE	TE	TE	TE	TE	PL	PL	PL	PL	CR	CR	CR	CR	CR	CR	CO
11	CR	CR	CR	CR	CR	CR	PL	PL	PL	PL	TE	TE	TE	TE	TE	TE	CO	CO	CO	CO	CR
12	CR	CR	CO	CO	CO	CO	TE	TE	TE	TE	TE	TE	PL	PL	PL	PL	CR	CR	CR	CR	CR

Odd zero-based rows (printed rows 2, 4, 6, ...) are worked back across (turned), so the fabric mirrors while the sequence keeps consuming in order. Row start offsets: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11.

Yardage, step by step

Example swatch: 10 rows use 6.5 yards.
Yards per row = 6.5 ÷ 10 = 0.65 yards/row.
Planned rows = 220 rows × 1 panel(s) = 220 rows.
Total stitches = 21 × 220 = 4620 stitches.
Total yarn = 0.65 × 220 = 143 yards.
With 20% overage = 143 × (1 + 20/100) = 171.6 yards.
Skeins = round up (171.6 ÷ 231) = **1 skein(s)** of 231 yards each.

Plaid Columns Blanket Panel

Plaid Columns Blanket Panel · Worsted (CYC 4) · single crochet

> Worked planning example: every stitch number below was computed with the FiberTools pooling model; the swatch yardage is an example measurement — replace it with your own swatch numbers.

Color	Name	Hex	Whole stitches
RU	Rust	#B3541E	8
GO	Gold	#D9A441	5
BR	Brown	#5C4033	8
OL	Olive	#707A3A	5

Total repeat: 26 stitches

Adjustment	Row width	Shift per row
-2	24	24
-1	25	25
0	26	0
1	27	1
2	28	2

Chosen adjustment: **0**; row width: 26; shift per row: 0. Shift 0 stacks colors into plaid columns in the sequence; turned rows mirror the fabric as shown below.

Expected pattern grid

1	RU	RU	RU	RU	RU	RU	RU	RU	GO	GO	GO	GO	GO	BR	BR	BR	BR	BR	BR	BR	BR	OL	OL	OL	OL	OL
2	OL	OL	OL	OL	OL	BR	BR	BR	BR	BR	BR	BR	BR	GO	GO	GO	GO	GO	RU	RU	RU	RU	RU	RU	RU	RU
3	RU	RU	RU	RU	RU	RU	RU	RU	GO	GO	GO	GO	GO	BR	BR	BR	BR	BR	BR	BR	BR	OL	OL	OL	OL	OL
4	OL	OL	OL	OL	OL	BR	BR	BR	BR	BR	BR	BR	BR	GO	GO	GO	GO	GO	RU	RU	RU	RU	RU	RU	RU	RU
5	RU	RU	RU	RU	RU	RU	RU	RU	GO	GO	GO	GO	GO	BR	BR	BR	BR	BR	BR	BR	BR	OL	OL	OL	OL	OL
6	OL	OL	OL	OL	OL	BR	BR	BR	BR	BR	BR	BR	BR	GO	GO	GO	GO	GO	RU	RU	RU	RU	RU	RU	RU	RU
7	RU	RU	RU	RU	RU	RU	RU	RU	GO	GO	GO	GO	GO	BR	BR	BR	BR	BR	BR	BR	BR	OL	OL	OL	OL	OL
8	OL	OL	OL	OL	OL	BR	BR	BR	BR	BR	BR	BR	BR	GO	GO	GO	GO	GO	RU	RU	RU	RU	RU	RU	RU	RU

9	RU	RU	RU	RU	RU	RU	RU	RU	GO	GO	GO	GO	GO	BR	BR	BR	BR	BR	BR	BR	BR	OL	OL	OL	OL	OL
10	OL	OL	OL	OL	OL	BR	BR	BR	BR	BR	BR	BR	BR	GO	GO	GO	GO	GO	RU	RU	RU	RU	RU	RU	RU	RU
11	RU	RU	RU	RU	RU	RU	RU	RU	GO	GO	GO	GO	GO	BR	BR	BR	BR	BR	BR	BR	BR	OL	OL	OL	OL	OL
12	OL	OL	OL	OL	OL	BR	BR	BR	BR	BR	BR	BR	BR	GO	GO	GO	GO	GO	RU	RU	RU	RU	RU	RU	RU	RU

Odd zero-based rows (printed rows 2, 4, 6, ...) are worked back across (turned), so the fabric mirrors while the sequence keeps consuming in order. Row start offsets: 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0.

Yardage, step by step

Example swatch: 10 rows use 11 yards.
Yards per row = 11 ÷ 10 = 1.1 yards/row.
Planned rows = 120 rows × 4 panel(s) = 480 rows.
Total stitches = 26 × 480 = 12480 stitches.
Total yarn = 1.1 × 480 = 528 yards.
With 15% overage = 528 × (1 + 15/100) = 607.2 yards.
Skeins = round up (607.2 ÷ 220) = **3 skein(s)** of 220 yards each.

Stair-Step Cowl

Stair-Step Cowl · DK (CYC 3) · half double crochet

> Worked planning example: every stitch number below was computed with the FiberTools pooling model; the swatch yardage is an example measurement — replace it with your own swatch numbers.

Color	Name	Hex	Whole stitches
NA	Navy	#1B2A6B	5
SK	Sky	#7FB2D9	3
WH	White	#F5F5F0	5
RE	Red	#B03A2E	3

Total repeat: 16 stitches

Adjustment	Row width	Shift per row
-2	14	14
-1	15	15
0	16	0
1	17	1
2	18	2

Chosen adjustment: **2**; row width: 18; shift per row: 2. A two-stitch step makes a faster stair-step that wraps the repeat every 8 rows.

Expected pattern grid

1	NA	NA	NA	NA	NA	SK	SK	SK	WH	WH	WH	WH	WH	RE	RE	RE	NA	NA
2	NA	NA	NA	NA	RE	RE	RE	WH	WH	WH	WH	WH	SK	SK	SK	NA	NA	NA
3	NA	SK	SK	SK	WH	WH	WH	WH	WH	RE	RE	RE	NA	NA	NA	NA	NA	SK
4	SK	SK	SK	NA	NA	NA	NA	NA	RE	RE	RE	WH	WH	WH	WH	WH	SK	SK
5	WH	WH	WH	WH	WH	RE	RE	RE	NA	NA	NA	NA	NA	SK	SK	SK	WH	WH
6	WH	WH	WH	WH	SK	SK	SK	NA	NA	NA	NA	NA	RE	RE	RE	WH	WH	WH
7	WH	RE	RE	RE	NA	NA	NA	NA	NA	SK	SK	SK	WH	WH	WH	WH	WH	RE
8	RE	RE	RE	WH	WH	WH	WH	WH	SK	SK	SK	NA	NA	NA	NA	NA	RE	RE

9	NA	NA	NA	NA	NA	SK	SK	SK	WH	WH	WH	WH	WH	RE	RE	RE	NA	NA
10	NA	NA	NA	NA	RE	RE	RE	WH	WH	WH	WH	WH	SK	SK	SK	NA	NA	NA
11	NA	SK	SK	SK	WH	WH	WH	WH	WH	RE	RE	RE	NA	NA	NA	NA	NA	SK
12	SK	SK	SK	NA	NA	NA	NA	NA	RE	RE	RE	WH	WH	WH	WH	WH	SK	SK

Odd zero-based rows (printed rows 2, 4, 6, …) are worked back across (turned), so the fabric mirrors while the sequence keeps consuming in order. Row start offsets: 0, 2, 4, 6, 8, 10, 12, 14, 0, 2, 4, 6.

Yardage, step by step

Example swatch: 10 rows use 5 yards.
Yards per row = 5 ÷ 10 = 0.5 yards/row.
Planned rows = 60 rows × 1 panel(s) = 60 rows.
Total stitches = 18 × 60 = 1080 stitches.
Total yarn = 0.5 × 60 = 30 yards.
With 20% overage = 30 × (1 + 20/100) = 36 yards.
Skeins = round up (36 ÷ 246) = **1 skein(s)** of 246 yards each.

Dye-Lot Buying Checklist

- [ ] Buy all the yarn for the project before you start — pooling eats the sequence in order and a mid-project reorder can change the repeat.
- [ ] Check the dye-lot number on every label matches; photograph the labels with the first swatch.
- [ ] Buy the swatch-math yardage plus the overage (15-20 percent, or one extra skein on narrow pieces) from the SAME lot.
- [ ] Compare each skein: unwind a few yards and confirm the color sequence and section lengths look the same before you wind it into your project.
- [ ] Keep one full label and the receipt with the project until it is finished.
- [ ] If a skein runs out mid-row, start the new skein at a color-section boundary, not mid-color.
- [ ] If you are forced to change dye lots, hide the change at a seam, an edge, or a panel break — expect the repeat lengths to differ slightly.

Troubleshooting Grid

Symptom	Likely cause	Fix
Pattern drifts diagonally but I wanted straight columns	Shift per row is not 0 — the row width is not the repeat	Set adjustment so row width equals the repeat exactly (shift 0).
Columns stack, then slowly creep sideways after many rows	Tension changes across the row or between sessions	Re-swatch at project width; keep the measuring routine identical; note the row where creep starts in the trial log.
Pattern breaks suddenly at one point in a row	A factory join interrupted the color sequence	Cut the yarn at the join and restart at the beginning of the color sequence.
New skein starts in the middle of a color	Skeins do not all start at the same sequence point	Wind off or cut forward to the start of a color section before joining.
My edges eat stitches the preview does not show	Setup and turning chains consume part of the repeat	The model assumes no chain consumption — measure sections in worked stitches only and treat chain stitches as outside the repeat.
One color band keeps getting wider or narrower	Repeat length varies between skeins or dye lots	Measure the new skein sections before joining; if they differ, reserve that skein for a separate panel.
Color boundaries look fuzzy or split	A boundary lands within a stitch instead of between stitches	Count sections in whole stitches only and re-measure at a clean boundary; the grid is idealized.
Pooling worked in the swatch but not in the piece	Gauge or tension at project width differs from the swatch	Swatch at the full project width; adjust hook size before adjusting the stitch counts.

From preview to finished piece

1. Measure one full color sequence in whole stitches and record your yarn and dye lot.
2. Try widths in the free calculator, then swatch at project width and log the result.
3. Measure swatch yardage, scale to all planned rows, add overage, and round up skeins from the same lot.
4. Track your rows and check tension, sequence boundaries, and each new skein as you work.

Whole stitches only; color boundaries can land inside a stitch. No foundation-chain or turning-chain consumption is modeled. Tension, joins, edges, and repeat variation change real fabric. The grid is an idealized planning aid, not a guaranteed pooling outcome.

The free Planned Color Pooling Calculator and all other FiberTools calculators are at fibertools.app — they stay free.